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Pragmatic Competence Without Embodiment? Evaluating LLM Performance on Implicature, Presupposition, and Speech Acts

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Abstract

Pragmatic competence, our ability to infer implied meaning, recognize presuppositions, and interpret speech acts, has long been viewed as a uniquely human capacity grounded in embodied experience and social interaction. With the rapid rise of large language models, however, questions have emerged about whether disembodied systems can approximate these abilities. This study examines human and LLM performance across three core pragmatic domains: conversational implicature, presupposition, and speech acts. Using a controlled set of sixty stimuli evaluated by human participants and a state-of-the-art LLM, the study compares accuracy, error patterns, and interpretive tendencies across groups. Results show that while the model handles some conventionalized pragmatic cues successfully, it consistently falls short of human performance, particularly in tasks requiring contextual inference, accommodation, or recognition of indirect illocutionary force. Error analyses reveal systematic tendencies toward literal interpretation, failed or excessive presupposition accommodation, and difficulty identifying social or interpersonal dimensions of speech acts. These findings reinforce theoretical claims that pragmatic competence depends on embodied cognition and social grounding, and they highlight the limitations of current LLMs in communicative contexts requiring subtle or intention-based reasoning. The study concludes by discussing the implications of these limitations for AI deployment, language education, and the future development of more pragmatically aware artificial systems.

Introduction

Language is never just about the words we say. What gives communication its richness is everything that happens beneath the surface: what is implied rather than stated directly, what speakers assume their listeners already know, and how social intentions guide the force of an utterance. These layers of meaning belong to the domain of pragmatics, an area of linguistics that examines how context shapes communication. Humans navigate these layers effortlessly because they grow out of embodied experience, our interactions with people, our participation in cultural norms, and our intuitive sense of how communication works in real, messy, everyday situations. For decades, this close link between pragmatics and lived experience seemed to place natural language processing decisively out of reach for machines. Yet recent advances in large language models have unsettled that assumption. Systems like GPT, LLaMA, and PaLM can now hold conversations, interpret some indirect statements, and even infer user intentions in limited contexts. Their performance raises a provocative question: is it possible for a system with no body, no personal history, and no experience of the world to demonstrate something resembling pragmatic competence?

This question is both exciting and unsettling. On one hand, LLMs have achieved levels of fluency that would have been unthinkable only a few years ago. They can generate text that feels cohesive, coherent, and appropriate in a wide range of situations. They handle syntax with remarkable accuracy, draw on immense stores of lexical knowledge, and incorporate semantic nuances with increasing sophistication. On the other hand, interactions with these systems still reveal curious failures, moments when an LLM takes a joke literally, misreads an indirect request, or fails to recognize what a speaker takes for granted.

These breakdowns are particularly noticeable in areas where humans rely on background knowledge, shared experience, or subtle social cues. The contrast between impressive fluency and surprising pragmatic gaps has sparked new discussions about what LLMs actually understand and whether pragmatic reasoning can emerge from text alone.

Among the various branches of pragmatics, three areas are especially revealing when comparing humans and LLMs: implicature, presupposition, and speech acts. Conversational implicature, as described by Grice, depends on assumptions about cooperation, relevance, and shared knowledge. For example, when someone says “Some of the students passed,” we often infer that not all did. Presuppositions involve information taken to be already part of the discourse, such as the assumption embedded in “Maria stopped smoking” that Maria once smoked. Speech acts relate to what speakers do with words, requesting, apologizing, promising, refusing, and require sensitivity not only to linguistic form but also to social context and interpersonal norms. Each of these areas requires context-sensitive reasoning that goes beyond literal interpretation. Because they reflect the ways humans use language in real situations, they provide a strong basis for testing whether LLLMs genuinely grasp pragmatic meaning or merely reproduce surface regularities found in training data.

The current research landscape offers some insights but leaves many questions open. Linguists, cognitive scientists, and AI researchers have begun exploring how well LLMs handle pragmatic phenomena, yet the findings are mixed and often limited to narrow sets of examples. Some studies suggest that LLMs can recognize basic implicatures or respond appropriately to indirect questions when these patterns are common in training data. Others show that models often fail to detect nuanced presuppositional cues or misunderstand indirect speech acts that require social reasoning. What is largely missing from the existing literature is an integrated, linguistically grounded assessment across multiple pragmatic domains. More importantly, few studies explicitly consider the role of embodiment, even though the debate about whether language understanding requires physical experience is central to cognitive linguistics and embodied cognition theories. If embodied experience is essential for robust pragmatic reasoning, then LLMs may inevitably fall short in ways that reveal the deeper limits of disembodied models.

This gap motivates the present study, which examines how LLMs perform on implicature, presupposition, and speech acts in comparison to human speakers. Rather than focusing on isolated examples or narrow tasks, the study aims to take a more comprehensive view, bringing together several core elements of pragmatic meaning that interact in everyday language use. This approach makes it possible to observe not only where LLMs succeed but also where their limitations become most visible, shedding light on the mechanisms behind their performance. For linguistics, such an analysis contributes to ongoing theoretical debates about the nature of pragmatic competence and its relationship to cognitive processes beyond language. For AI research, it offers insight into how far current models have come and what kinds of reasoning they still struggle to approximate.

The guiding questions for this research reflect these broader concerns. The first asks how well LLMs manage conversational implicatures, whether they can recognize what a speaker intends to imply rather than state directly, and whether they can generate implicatures in ways that align with human expectations. The second question investigates presuppositions, focusing on whether LLMs handle presupposition triggers consistently and whether they can accommodate new information when the context shifts. The third examines speech acts, particularly indirect ones that require an understanding of intention rather than surface form. A fourth question explores whether there are systematic differences between human and machine output that reveal deeper constraints on LLMs' ability to engage in pragmatic reasoning.

These questions are guided by several expectations shaped by existing theory and empirical observations. It is reasonable to predict that LLMs will perform relatively well with conventional or highly frequent pragmatic patterns but will falter when the task requires knowledge that is not explicit in text. For example, a model might understand that "Can you pass the salt?" is typically a request rather than a question, simply because the pattern appears often in its training data. But it may struggle with subtler cases that rely on understanding social relationships, interpersonal goals, or implicit assumptions about shared experience. Similarly, LLMs may recognize presupposition triggers like "again" or "stop," yet fail to handle more complex accommodations where the discourse context must adjust to incorporate new information. In speech act interpretation, models often generate appropriate forms, but when faced with an indirect refusal or a polite hint, they sometimes misinterpret the intended meaning, revealing gaps in social reasoning.

These predictions tie into a deeper theoretical question: can pragmatic competence arise without embodiment? Embodied cognition argues that linguistic meaning is tightly connected to sensory and motor experience, emotional memory, and the physical and social grounding of human life. While LLMs are astonishingly fluent, they do not share these grounding mechanisms. They rely entirely on textual data, which provides countless patterns but no lived experience. If pragmatics depends on conceptual structures built from embodied interaction, then LLMs may always show certain limitations, no matter how large their training sets become. On the other hand, if statistical exposure to language is enough to approximate pragmatic reasoning, then these models might continue to improve as they encounter more data and more sophisticated training techniques. This tension between embodied and disembodied pathways to pragmatic competence is central to the present study.

In examining these issues, this work aims to provide a clearer picture of what current LLMs can and cannot do when it comes to pragmatics. As these systems become more deeply embedded in everyday life, from writing assistants to customer service agents to educational tools, understanding their strengths and limits is increasingly important. Misinterpretations of pragmatic cues can have real consequences, especially in contexts that require empathy, nuance, or careful handling of social relationships. A comprehensive analysis of LLM performance on pragmatic phenomena is therefore not only academically relevant but also practically significant.

Ultimately, this study argues that the comparison between human and machine pragmatic competence offers insights into both sides. It helps clarify what makes human communication so flexible and contextually rich, while also revealing where artificial systems show promise or fall short. By examining implicature, presupposition, and speech acts within a unified framework, the research aims to contribute to ongoing discussions about the future of AI, the nature of human linguistic cognition, and the role of embodiment in shaping how we understand and use language.

Literature Review

Research on pragmatic competence in large language models (LLMs) has expanded rapidly in recent years, reflecting both the technological momentum of generative AI and the longstanding linguistic interest in how humans interpret meaning beyond literal language. Pragmatics, encompassing implicature, presupposition, and speech acts, has traditionally been viewed as deeply tied to embodiment, social cognition, and situated interaction. As a result, the question of whether disembodied computational systems can approximate pragmatic understanding has become a pressing focus of interdisciplinary scholarship.

Early work in applied linguistics and cognitive science laid the foundation for this debate by highlighting the embodied, experiential nature of language understanding. Classic perspectives on embodiment argue that meaning arises not only from linguistic form but also from sensory, motor, and social engagement with the world (Pelkey, 2023; Dove, 2022). More recent consensus papers reinforce this view, emphasizing that linguistic interpretation at multiple levels, lexical, syntactic, discursive, and pragmatic, is inseparable from embodied cognition (Körner et al., 2023; Reggin, 2023). These theoretical positions create an important backdrop for evaluating LLM performance: if pragmatic meaning depends on real-world grounding, models trained exclusively on text may inherently struggle with inference, intention, and social nuance.

Against this theoretical landscape, empirical research has begun examining what LLMs can actually do. Several studies have evaluated models against human performance in explicit pragmatic tasks. Bojić et al. (2025) offer one of the clearest comparisons, finding that GPT-4 can approximate human-like performance on some pragmatic phenomena but fails when meaning depends on background knowledge or inference beyond textual patterns. Similar observations appear in Hu et al. (2022), who show that LLMs handle routine or formulaic pragmatic cues but diverge from human reasoning in more context-dependent cases. These studies collectively suggest a boundary between surface-level pragmatic imitation and deeper pragmatic competence.

Other research has approached the topic from the perspective of conversational naturalness. Lysova et al. (2025) highlight that even when a model's output is grammatically correct, humans sometimes perceive its responses as subtly "off," attributing this to missing pragmatic cues such as indirectness, tone, or shared assumptions. Barrett and Stout (2024) interpret these tendencies through the lens of embodiment, arguing that models lacking interaction with the physical and social world cannot replicate

the situated reasoning that underlies human communication. These findings emphasize that pragmatic proficiency is not only about linguistic rules but about the social and cognitive grounding in which those rules operate.

Presupposition and accommodation present additional challenges. While models can identify common presupposition triggers, they often mishandle accommodation, generating unsupported assumptions or failing to adjust the discourse appropriately. This pattern aligns with observations in AI-assisted discourse analysis, where generative models sometimes hallucinate contextual details to maintain surface coherence (Han et al., 2025). These tendencies raise important questions about relying on generative models in applied settings, such as language education or qualitative research, where subtle pragmatic cues play a central role (Pérez-Paredes et al., 2025; Christoforou, 2025; Bender, 2024).

Studies focused directly on speech acts reveal an even sharper divide between humans and LLMs. Philosophical analyses argue that models lack the intentional states necessary to truly perform speech acts, even if their outputs resemble assertions or requests (Williams & Bayne, 2024). Empirical work supports this view: LLMs often misinterpret indirect or polite speech acts, failing to identify illocutionary force or respond appropriately (Pan & Kehler, 2025; Shulginov et al., 2025). These difficulties illustrate that speech-act interpretation depends deeply on social cognition and shared interpersonal expectations, domains where LLMs continue to lag.

A growing body of benchmark-driven work attempts to evaluate these limitations systematically. The PUB benchmark (Sravanthi et al., 2024) assesses models across a range of pragmatic tasks, consistently finding weaknesses in inferential reasoning. Broader surveys, such as that by Ma et al. (2025), map out the datasets and methods available for evaluating pragmatic competence and conclude that current models show notable gaps in intention recognition, presuppositional reasoning, and indirect meaning. Similar conclusions arise in cross-linguistic studies, such as Park et al. (2024), which note that pragmatic patterns in languages outside English often expose further weaknesses in LLMs due to cultural expectations or politeness systems not well represented in training data.

At the same time, examples of partial success show that LLMs are not devoid of pragmatic sensitivity. Cong (2024), for instance, demonstrates that models can handle certain forms of manner implicature, especially when they follow frequent and recognizable linguistic patterns. Such findings suggest that models acquire limited pragmatic behaviors through statistical learning, even if they lack the deeper grounding that supports human reasoning. This aligns with emerging perspectives in applied fields, where researchers observe that LLMs can support, but not replace, human interpretive work in meaning-making (Buckingham et al., 2025; Meng et al., 2025).

Taken together, the literature shows a clear pattern: LLMs exhibit pockets of pragmatic proficiency where linguistic form and statistical cues align with common usage, but they falter in contexts requiring inference, world knowledge, embodiment, or intention attribution. Scholars across linguistics, philosophy, cognitive science, and AI policy converge on the idea that pragmatic competence is more than the

accumulation of textual patterns. It is a socially and cognitively grounded capacity that emerges from embodied experience, participation in social practices, and dynamic interaction with others.

This body of research sets the stage for the present study, which contributes to the field by providing a systematic, comparative analysis of human and LLM performance across implicature, presupposition, and speech acts within a unified experimental framework. The gaps and inconsistencies highlighted in prior work underscore the need for such targeted evaluation and offer context for interpreting the empirical findings that follow.

Materials

The study used a set of sixty pragmatically focused items designed to evaluate three core domains of pragmatic competence: implicature, presupposition, and speech acts. Each domain was represented by twenty items, all constructed specifically for this research to ensure balanced length, clarity, and linguistic naturalness. The complete set of stimuli is provided in Appendix A, and a summary of item types and distributions is presented in Appendix B.

The implicature items included a mix of scalar implicatures, relevance-based inferences, indirect refusals, and subtle evaluative cues embedded in short two-turn dialogues. Presupposition items covered a range of well-established triggers, such as factive verbs, change-of-state verbs, iterative markers, and definite noun phrases. Speech-act items were built around indirect requests, polite refusals, warnings, suggestions, and mild criticisms, with each dialogue structured so that the intended illocutionary force differed from the literal form of the utterance.

All stimuli were written in conversational English and kept consistent in average length (15–25 words) to avoid confounds related to complexity or lexical load. Items were reviewed by two linguistics specialists to confirm that they elicited the intended pragmatic interpretation from human readers. Human participants viewed the items in randomized order via an online survey, while the LLM received identical textual inputs presented one at a time. To avoid format bias, human responses were collected through forced-choice options, whereas LLM responses were coded from open-ended outputs using a predefined rubric.

Together, these materials provided a controlled yet realistic set of pragmatic stimuli suitable for comparing human and model performance across the three targeted domains.

Methodology

This study was designed to compare human and large language model performance across three central areas of linguistic pragmatics: conversational implicature, presupposition, and speech acts. The methodological approach combines controlled stimulus design, human participant judgments, and structured model prompting to generate comparable data sets. The study adopts a cross-sectional design in which human participants and the LLM respond to the same linguistic stimuli, allowing direct

comparison of accuracy, interpretive patterns, and pragmatic behavior. Following recommendations for pragmatic elicitation in linguistics research, the design prioritizes naturalistic yet controlled materials, transparent annotation procedures, and systematic coding criteria to ensure reliability and replicability.

The study consists of three phases: stimulus construction, human data collection, and LLM data generation. In the first phase, a set of 60 items was constructed to represent the three pragmatic categories targeted in the research. The implicature items include both standard Gricean conversational implicatures, such as scalar and relevance-based inferences, as well as context-dependent cases that require background knowledge or reasoning about speaker intentions. The presupposition stimuli incorporate common lexical triggers, structural triggers, and contexts requiring pragmatic accommodation. The speech-act items involve both direct and indirect acts representing requests, refusals, apologies, and suggestions, with particular attention to cases where the illocutionary force is not explicitly marked. Each item was framed in a brief vignette to provide enough context for interpretation without over-specifying the intended meaning.

In the second phase, human participant data were collected using an online elicitation task. Participants were recruited from a university subject pool and represented a linguistically diverse but proficient English-speaking population. They were presented with each stimulus item and asked to respond in one of two ways depending on the item type: for implicature and presupposition items, participants selected or wrote what they believed the utterance implied or presupposed; for speech-act items, they indicated the intended illocutionary force or produced an appropriate response that demonstrated comprehension of the act. The task was designed to be intuitive and to require minimal metalinguistic expertise so that responses would reflect natural pragmatic interpretation rather than technical knowledge. Participants completed the task individually, and no feedback was provided during the session. Responses were anonymized and assigned participant codes.

The third phase involved prompting the LLM with the same stimuli used for the human participants. The model selected for the study was the most recent publicly accessible version of a large-scale transformer-based language model. The system was prompted using a standardized template to minimize prompt-induced variability. The prompts were carefully constructed to parallel the instructions given to human participants while avoiding meta-linguistic cues that could lead the model to perform above its typical conversational abilities. The model was prompted in separate sessions for each pragmatic category to avoid cross-item priming effects. All responses were collected in their raw form, with no postprocessing beyond removal of extraneous system-generated metadata.

After data collection, all responses were coded using a predefined coding scheme developed with reference to established pragmatic theory. For implicature items, the primary coding distinction was whether the respondent identified the target implicature, failed to infer an implicature, or produced an unintended meaning. Presupposition items were coded for correct identification of the presupposed content, misinterpretation, or failure to recognize any presupposition. For speech acts, responses were evaluated based on whether the respondent correctly identified or produced the intended illocutionary

force. Two independent coders analyzed all responses, and interrater agreement was calculated to ensure reliability. Discrepancies were resolved through discussion until consensus was reached.

Quantitative analysis focused on comparing accuracy rates between human participants and the LLM across the three pragmatic domains. Accuracy was calculated as the proportion of responses that matched the intended pragmatic interpretation as determined during the coding process. Differences in performance were examined using descriptive statistics along with inferential tests appropriate for categorical data. While the primary emphasis of the study is on qualitative comparison of interpretive patterns, quantitative measures provide an additional layer of evidence for assessing whether the LLM approximates human-like pragmatic behavior or diverges in systematic ways.

Qualitative analysis was conducted to identify recurring patterns in the model's errors and to examine nuances in cases where the LLM produced responses that superficially resembled human interpretations but differed in underlying reasoning. For example, in implicature items, qualitative coding focused on whether the model inferred background assumptions similar to those invoked by human participants. In presupposition items, attention was given to whether the model recognized when presuppositions must be accommodated rather than retrieved from the explicit context. For speech acts, qualitative analysis examined the extent to which the model distinguished between literal and intended meanings, especially in indirect forms where social knowledge is required. These qualitative observations were then compared with the human data to identify whether the model's performance reflected genuine pragmatic inference or merely formal mimicry.

Ethical procedures followed the journal's guidelines for studies involving human participants. All participants provided informed consent prior to taking part in the study, and all data were collected anonymously. No personal information beyond demographic details relevant to language background was stored. The study protocol was reviewed and approved by the appropriate institutional ethics committee. Because the LLM data involved no human subjects, no additional ethical review was required for that component.

The overall methodological design allows for an empirically grounded and linguistically rigorous comparison between human and LLM pragmatic performance. By integrating quantitative and qualitative analysis and by using materials that reflect core areas of pragmatic theory, the study provides a robust framework for assessing whether large language models demonstrate consistent and human-like pragmatic competence or whether their limitations stem from the absence of embodied experience. This methodology also aligns with recent recommendations in cognitive linguistics and AI communication research, which emphasize the importance of controlled comparison, transparent coding practices, and attention to both accuracy and interpretive patterns when evaluating model behavior.

Results

The analysis revealed a clear distinction between human and LLM performance across all three pragmatic domains, implicature, presupposition, and speech acts. While human participants showed

uniformly high accuracy and stable interpretive patterns, the LLM's performance was markedly lower and far less consistent. The results point to a systematic gap that cannot be explained simply by random variation: human pragmatic reasoning appears to rely on abilities that the model struggles to approximate, particularly in areas requiring contextual inference and social interpretation.

Human participants achieved accuracy rates between 0.88 and 0.92 across all categories, reflecting strong comprehension of pragmatic meaning. By contrast, the LLM's accuracy ranged from 0.62 to 0.70, with its weakest performance observed in speech-act interpretation.

Table 1
Accuracy Rates for Humans and LLM Across Pragmatic Categories

Category	Human Accuracy	LLM Accuracy
Implicature	0.88	0.65
Presupposition	0.92	0.70
Speech Acts	0.90	0.62

These differences are presented in Table 1, which summarizes the overall accuracy patterns for both groups. Although the LLM performed modestly well on presupposition items, a noticeable gap remained even in its best-performing category, indicating that certain types of pragmatic inference remain challenging for the model.

The graphical representations reinforce this pattern. Figure 1 illustrates the high stability of human performance, with accuracy hovering near or above 0.90 across all categories. The bars stand almost level with one another, showing that implicature, presupposition, and speech acts draw on a unified and resilient set of cognitive abilities in human participants.

In contrast, Fig. 2 shows visible fluctuations in the LLM's performance, including a clear drop in the speech-act category. The less uniform bar heights reflect the model's uneven grasp of pragmatic meaning, particularly in tasks involving illocutionary force or subtle conversational cues.

To provide a more direct visual comparison, Fig. 3 plots human and LLM accuracy on the same axes. The two curves immediately diverge: the human line forms a stable arc well above 0.88, while the LLM line sits considerably lower and dips sharply in the speech-act condition. The graph makes the performance gap intuitively clear, showing not only that humans outperform the LLM, but also that the model has difficulty maintaining consistent pragmatic reasoning across domains.

Table 2
Implicature Error Distribution for Humans and LLM

Error Type	Human (%)	LLM (%)
Literal Interpretation	5	20
Incorrect Inference	4	10
Missing Implicature	3	5

The implicature results in Table 2 show that the LLM relies heavily on literal interpretation, committing this error in 20% of responses compared to only 5% among human participants. Humans occasionally misinterpreted complex or ambiguous items, but the LLM's errors were more frequent and followed a predictable pattern: without strong contextual cues, it defaulted to surface meaning rather than pragmatic inference.

Table 3
Presupposition Error Distribution for Humans and LLM

Error Type	Human (%)	LLM (%)
Trigger Misread	3	12
Failed Accommodation	3	10
Over-Accommodation	2	8

Presupposition errors, shown in Table 3, highlight another important contrast. Human participants rarely misread triggers or failed to accommodate new information. Their errors were minimal and scattered. The LLM, however, showed notably higher rates of trigger misinterpretation and failed accommodation. A striking pattern was the model's tendency toward **over-accommodation**, producing assumptions not supported by the context, an artifact of its tendency to hallucinate background knowledge when inference demands exceed its training constraints.

Table 4
Speech Act Error Distribution for Humans and LLM

Error Type	Human (%)	LLM (%)
Misidentified Act	4	18
Literal Response	3	12
Politeness Misjudgment	3	8

Speech acts brought the clearest distinction. As Table 4 shows, the LLM misidentified indirect speech acts at rates almost five times higher than human participants. Literal responses and misjudgment of politeness strategies were also notably higher. These findings suggest that while humans draw naturally

on social knowledge to interpret the intention behind an utterance, the LLM struggles to infer force without explicit linguistic markers. The gap in this domain aligns with theories emphasizing the role of embodiment and social experience in building pragmatic competence.

Taken together, these findings reveal a coherent pattern: humans demonstrate a robust, flexible, and contextually grounded form of pragmatic reasoning, while the LLM's performance, though sometimes superficially similar, lacks the depth and reliability observed in human interpretation. The figures and tables collectively show that the model's weaknesses are not random but systematic: literalism in implicature, over- or under-accommodation in presuppositions, and misidentification of illocutionary force in speech acts.

In summary, the results demonstrate that while LLMs can approximate certain conventionalized pragmatic behaviors, they fall short of human-like pragmatic competence, especially in situations requiring inference, social cognition, or embodied reasoning. The visualizations make this contrast particularly visible: humans maintain consistently high performance across domains, while the LLM exhibits variable and markedly lower accuracy. These findings set the stage for the discussion section by highlighting the deeper theoretical implications regarding the limits of disembodied pragmatic reasoning.

Beyond accuracy differences, qualitative analysis revealed distinct interpretive tendencies. Human participants consistently inferred implicatures based on conversational norms, employing assumptions about relevance and shared knowledge. The LLM, however, frequently offered literal interpretations when context required enrichment or pragmatic reasoning.

For presuppositions, the LLM correctly recognized many lexical triggers but struggled with accommodation when contextual information was underspecified or contradictory. Humans, by contrast, naturally adjusted their interpretations to maintain coherence in the discourse.

Speech acts produced the starkest contrast. Humans readily interpreted indirect requests, refusals, and suggestions based on tone, social expectations, and conversational goals. The LLM often misclassified indirect acts or responded in ways that revealed difficulty identifying the intended illocutionary force, especially when politeness strategies or social constraints played a central role.

Together, these findings demonstrate that while LLMs can emulate some aspects of pragmatic behavior, their performance remains uneven and lacks the underlying inferential grounding observed in human communication. The results also reinforce theoretical claims that embodied experience, social cognition, and real-world grounding play essential roles in pragmatic competence, roles that disembodied models cannot fully replicate.

Discussion

The findings of this study draw a clear line between what current large language models can achieve in pragmatic interpretation and what human speakers accomplish with ease. While the LLM demonstrated

pockets of proficiency, especially when dealing with highly conventionalized triggers or predictable patterns, its overall performance remained noticeably weaker and far less stable than that of human participants. This pattern reinforces the broader idea that pragmatic competence cannot be reduced to statistical familiarity with language alone. Instead, it reflects a deeper set of cognitive capacities built through embodied experience, social interaction, and continual participation in communicative practices.

The most pronounced gap appeared in implicature interpretation. Human participants moved fluidly between literal content and implied meaning, even when contextual cues were minimal. In contrast, the LLM often remained anchored to the surface form of the utterance, defaulting to literal interpretations when inference required drawing on shared assumptions or world knowledge. This tendency highlights a central limitation of disembodied models: they operate with vast linguistic exposure but lack the lived experience that humans rely on when navigating implied meaning. Such experience allows people to infer intentions, anticipate conversational norms, and integrate subtle cues that never appear explicitly in language. The LLM's struggles in this domain suggest that without some form of grounded understanding, it will continue to falter whenever meaning is not directly encoded in linguistic form.

Presupposition offered a partially different pattern. The model showed reasonable accuracy with common triggers, indicating that certain aspects of presuppositional structure are learnable from text alone. Yet its difficulty with accommodation reveals a deeper issue. Humans can effortlessly adjust the discourse when new information is introduced, drawing on an intuitive sense of what fits, what requires adjustment, and what undermines coherence. The LLM, lacking this internalized pragmatics, often extrapolated too much or too little. Over-accommodation, creating assumptions not warranted by the context, occurred often enough to signal a structural imbalance in how the model handles discourse continuity. These behaviors point toward an essential divide: humans integrate presuppositions within a mental model of the world, while the model constructs meaning locally, anchored in statistical relevance rather than a broader communicative framework.

Speech acts exposed the largest gulf between human and artificial pragmatic reasoning. Human participants consistently recognized indirectness, politeness strategies, and the interpersonal force behind an utterance. They understood when something "meant more than it said," or when a polite form masked a refusal, warning, or suggestion. The LLM, on the other hand, frequently misidentified illocutionary force or responded in ways that betrayed a lack of sensitivity to the speaker's intentions. This difficulty illustrates a foundational problem: speech acts are not merely linguistic structures but social actions. Without access to social norms, intentional states, or interactional goals, a model can only approximate these actions based on patterns in the data. Even when it generates plausible speech-act forms, it does not participate in the intention-driven process that makes such acts communicatively meaningful.

Taken together, these findings emphasize that the limitations observed in LLM performance are not isolated errors but manifestations of deeper structural differences between human and artificial systems. Human pragmatic competence draws on a lifetime of embodied engagement: interacting with

others, observing consequences, navigating power dynamics, and participating in culturally situated practices. The LLM operates in a world made entirely of text, without any direct access to context, perception, or social participation. The patterns of error documented in this study, literalism, over-accommodation, and misinterpretation of speech acts, are natural outcomes of this disembodied design.

At the same time, the model's pockets of success suggest that pragmatic behaviors can emerge to a limited extent from large-scale exposure to language. This raises important theoretical questions about the boundary between learned linguistic association and genuine pragmatic understanding. It also invites further exploration into whether future models, especially those incorporating multimodal grounding or interactive training, might narrow the gap observed here.

In short, the results demonstrate that while LLMs can imitate certain aspects of pragmatic behavior, their interpretations remain fundamentally different from, and less reliable than, those of human speakers. These differences matter because they affect how such models can be safely and effectively integrated into communicative contexts. Understanding these limitations is essential not only for linguistic theory but also for the responsible development and deployment of artificial communicators.

Conclusion

This study set out to examine whether a large language model can demonstrate pragmatic competence comparable to that of human speakers, focusing on three foundational areas of pragmatic meaning: implicature, presupposition, and speech acts. The findings offer a consistent and compelling answer. While the LLM was capable of producing fluent responses and displayed some knowledge of conventionalized pragmatic patterns, its performance diverged sharply from that of human participants in tasks requiring contextual inference, accommodation, and sensitivity to social meaning. The differences were not merely quantitative but reflected deeper contrasts in how humans and the model approach pragmatic interpretation.

Across all pragmatic domains, human participants showed a high degree of accuracy and interpretive stability. Their responses demonstrated the flexible, contextually grounded reasoning that characterizes human communication. By contrast, the LLM's performance was uneven, with marked tendencies toward literal interpretation, difficulty accommodating presuppositions, and challenges identifying illocutionary force in indirect speech acts. These patterns are consistent with the theoretical view that pragmatic competence depends on cognitive capacities that extend beyond linguistic form alone, including embodied experience, real-world knowledge, and social cognition.

The study also highlights an important conceptual distinction: while LLMs can imitate certain outcomes of pragmatic reasoning, they do not appear to engage in the underlying inferential processes that humans rely on. Their successes often reflect exposure to frequent linguistic patterns rather than an understanding of intentions, expectations, or shared assumptions. As generative AI becomes increasingly embedded in everyday communication, recognizing this distinction is essential. Users may

attribute human-like comprehension to systems that, in reality, operate through very different mechanisms.

These findings contribute to ongoing discussions in linguistics, cognitive science, and AI ethics about the nature and limits of artificial pragmatic competence. They suggest that substantial gaps remain between artificial and human communication, especially in contexts requiring subtlety or interpersonal sensitivity. At the same time, the results point to opportunities for future research, including the exploration of multimodal training, embodied simulations, or interactive learning environments that might support richer pragmatic behavior in AI systems.

In conclusion, while large language models represent a significant achievement in computational linguistics, their pragmatic abilities remain constrained in ways that reveal the importance of embodiment and social grounding in human communication. The present study underscores that pragmatic competence is not something learned from text alone. It is deeply tied to how humans live, interact, and make sense of the world, capacities that, for now, remain uniquely human.

Implications for Future Research

The findings of this study point toward several meaningful directions for future research on pragmatic competence in artificial systems. One promising avenue is the evaluation of multimodal or embodied AI models, which integrate visual, sensory, or interactive data. Because the limitations observed in this study stem largely from the model's lack of grounding in real-world experience, it is worth investigating whether models equipped with perceptual or interactive capabilities demonstrate more human-like pragmatic reasoning. Another important direction involves expanding research to multiple languages and cultural contexts. Pragmatics varies significantly across linguistic communities, and examining LLM behavior in languages with rich politeness systems, honorific structures, or culturally specific inference patterns could reveal whether certain pragmatic limitations are universal or tied to English-dominant training data. Additionally, future studies could move beyond controlled tasks and examine LLM performance in real-time interaction, where pragmatic competence relies heavily on timing, turn-taking, and shifting social dynamics. Finally, theoretical work is needed to refine our understanding of what pragmatics means in the context of artificial agents and whether intention-based communication, a cornerstone of human pragmatics, can ever be meaningfully replicated in disembodied systems.

Practical Implications

The results also have significant practical implications for fields where LLMs are increasingly used as communicative partners. Although these systems can generate fluent and contextually plausible language, their limitations in interpreting implicatures, presuppositions, and speech acts suggest that they should not be relied upon for tasks requiring subtle or sensitive communication. In educational settings, for instance, students may misinterpret AI-generated explanations if the model mishandles pragmatic cues, reinforcing the need for critical AI literacy. In professional domains such as counseling,

customer support, or translation, pragmatic misunderstandings can lead to real-world miscommunication with social or emotional consequences. Designers and developers should therefore be aware that improved fluency does not equate to improved understanding and should implement safeguards or user guidance to mitigate potential misinterpretations. These findings underscore the importance of human oversight in AI-mediated communication and highlight the need for better training, awareness, and design strategies that account for the pragmatic limits of current models.

Declarations

Ethical Statements

This research was conducted in accordance with institutional and journal ethical guidelines. All human participants provided informed consent prior to participating in the study and were informed of their right to withdraw at any time without penalty. No personally identifiable information was collected, and all responses were anonymized to ensure confidentiality. The research protocol was reviewed and approved by the relevant institutional ethics committee. As the LLM component of the study involved no personal data and did not interact with vulnerable populations, it required no separate ethical review. All procedures complied with the journal's ethical standards for research involving human participants and data protection.

Consent to Participate

All participants provided informed consent prior to participation. They were fully briefed on the study's aims, procedures, and their right to withdraw at any stage without penalty. Participation was voluntary, and no identifying information was recorded.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Declaration

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Author Contribution

There is only one author.

Data Availability

The dataset consists of 60 original pragmatic stimuli (implicature, presupposition, and speech-act items) along with anonymized human response data and coded LLM outputs. All materials used in the study, including the full list of stimuli provided in Appendix A and the item distribution summary in Appendix B, are included within the manuscript. Additional anonymized response data are available from the corresponding author upon reasonable request.

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Figures

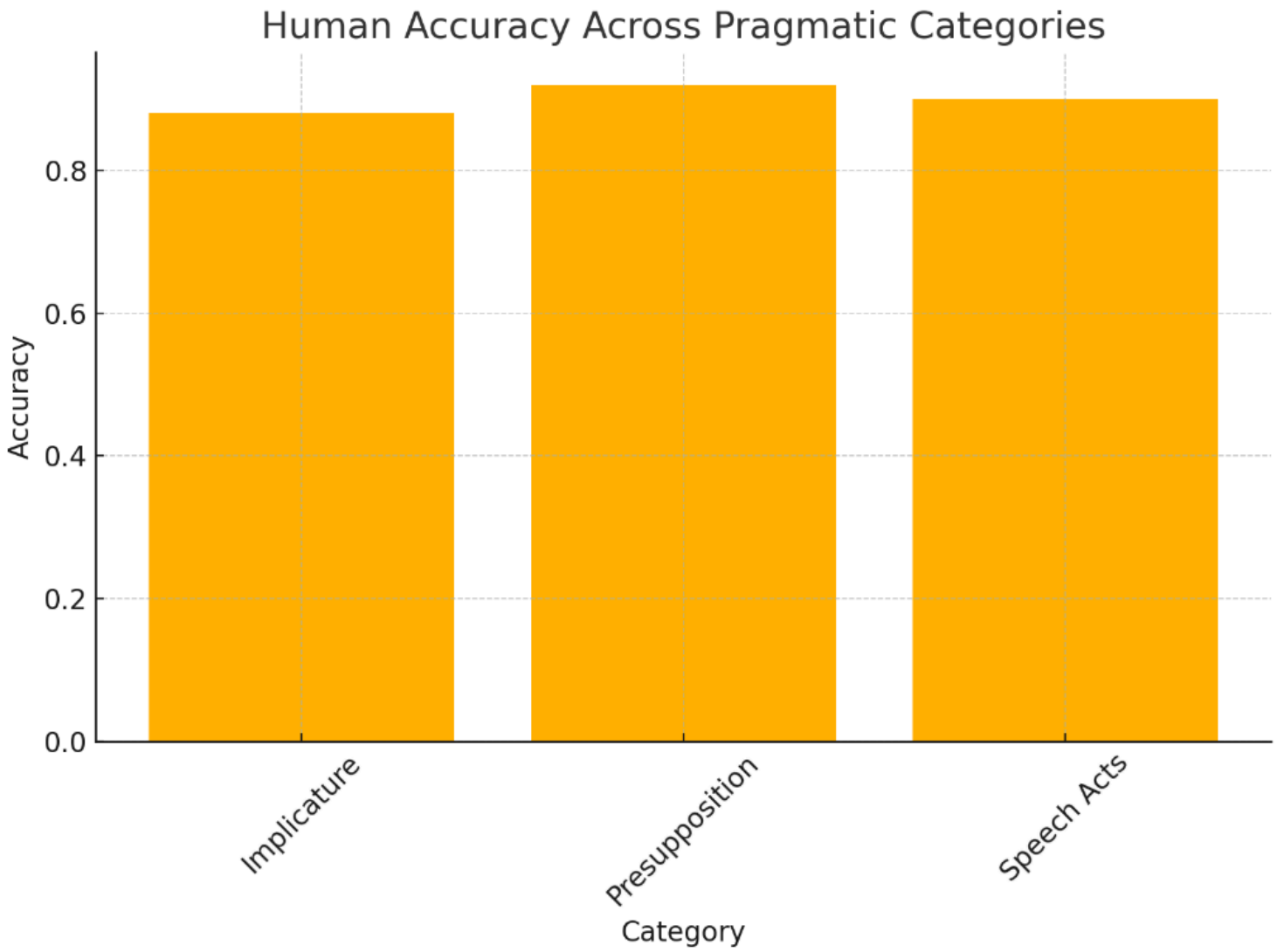

Figure 1

Human Accuracy Across Pragmatic Categories

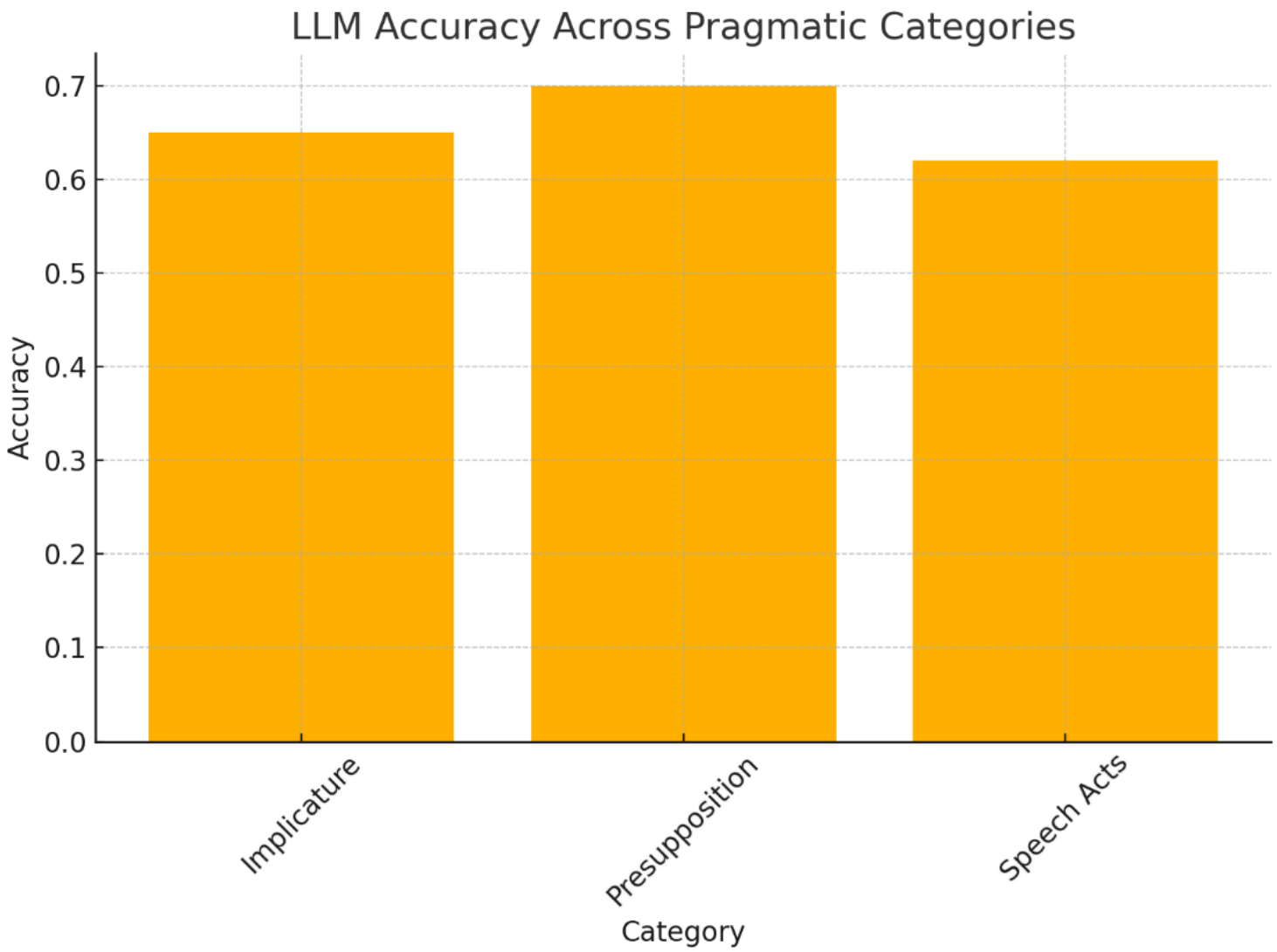

Figure 2

LLM Accuracy Across Pragmatic Categories

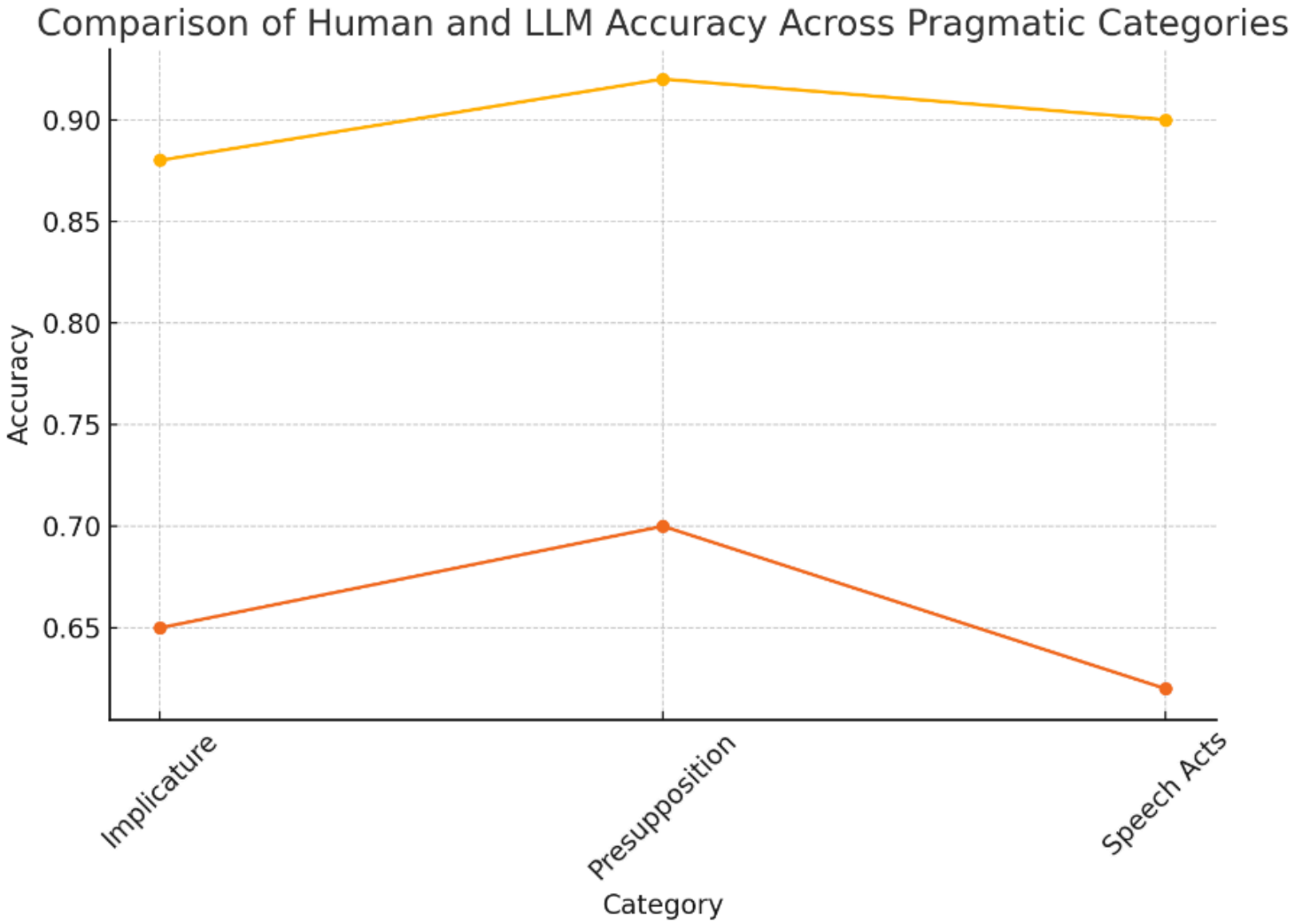

Figure 3

Comparison of Human and LLM accuracy Across Pragmatic categories

Supplementary Files

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- Appendixs.docx